HP Audio Player — Feature Migration Checklist

This document contains the complete feature inventory of the current HP Audio Player, categorized by module and complexity. Use this to track migration status.

1. Core Playback Engine

Status	#	Feature	Complexity	Source
[ ]	1.1	**HLS Streaming** (`.m3u8`) via hls.js	High	`use-hls-stream.ts`
[ ]	1.2	**Direct MP3/MP4/WAV** playback	Low	`simple-audio-player.tsx`
[ ]	1.3	**Auto-format detection** (HLS vs direct)	Low	`utils/formatters.ts`
[ ]	1.4	Play / Pause / Stop	Low	`playerSlice.ts`
[ ]	1.5	Seek (click on progress bar + keyboard)	Medium	`seek()` , `handleProgressBarClick()`
[ ]	1.6	Skip forward/backward (±10s/30s)	Low	`skip()`
[ ]	1.7	Volume control (slider + mute toggle)	Low	`changeVolume()` , `toggleMute()`
[ ]	1.8	Playback speed (0.5x → 2x)	Low	`changePlaybackRate()`
[ ]	1.9	Loop / Repeat single track	Low	`setIsLooping()`
[ ]	1.10	Buffer progress tracking	Low	`updateBufferProgress()`
[ ]	1.11	Error handling & retry logic	Medium	`onError()` , `retry()`
[ ]	1.12	Restart from beginning	Low	`restart()`

2. Player UI Modes

Status	#	Feature	Complexity	Source
[ ]	2.1	**Compact Player** (bottom bar)	High	`CompactPlayer.tsx` (10.5 KB)

[ ]	2.2	**Expanded Player** (full view with artwork)	Very High	`ExpandedPlayer.tsx` (82 KB!)
[ ]	2.3	**Mini Player** (floating pill)	Medium	`MiniPlayer.tsx` (13 KB)
[ ]	2.4	Mode switching (compact ↔ expanded ↔ mini)	Medium	`uiSlice.ts` , `toggleMiniPlayer()`
[ ]	2.5	Framer Motion animations between modes	Medium	`simple-audio-player.tsx`
[ ]	2.6	Responsive container dimensions tracking	Low	`ContainerDimensions` in `uiSlice.ts`

3. UI Controls & Interactions

Status	#	Feature	Complexity	Source
[ ]	3.1	Progress bar with hover time tooltip	Medium	`handleProgressBarClick()` , `hoverTime`
[ ]	3.2	Skip indicator visual feedback	Low	`SkipIndicator` in `uiSlice.ts`
[ ]	3.3	**Keyboard shortcuts** (Space, arrows, M, L, etc.)	Medium	`handleKeyDown()`
[ ]	3.4	**Drawer panel** (chapters / transcript / playlist tabs)	High	`DrawerTab` , `showDrawer`
[ ]	3.5	Idle timer (auto-hide controls)	Low	`use-idle-timer.tsx`
[ ]	3.6	Hover-to-show controls	Low	`isHovering` state
[ ]	3.7	**Mobile gestures** (swipe, touch)	Medium	`use-mobile-gestures.ts` (5.6 KB)
[ ]	3.8	Mobile optimization (orientation, battery)	Low	`use-mobile-optimization.ts`
[ ]	3.9	Mobile device detection	Low	`use-mobile.tsx`

4. Authentication & Security

Status	#	Feature	Complexity	Source

[ ]	4.1	**Password-protected streams**	High	`use-auth-session.ts` (24.8 KB)
[ ]	4.2	Session management (1-hour expiry)	Medium	`SESSION_DURATION` , `checkSessionExpiry()`
[ ]	4.3	**Signed URL authentication** (signature + ivhex)	High	`authSlice.ts` (15.5 KB)
[ ]	4.4	Password dialog UI	Medium	`showPasswordDialog` , popup auth flow
[ ]	4.5	Session persistence (save/clear)	Medium	`authSlice.ts`
[ ]	4.6	**Content protection** (anti-piracy)	High	`use-content-protection.ts` (13 KB)
[ ]	4.7	— DevTools detection	Medium	`detectDevTools()`
[ ]	4.8	— Right-click / copy / cut prevention	Low	`handleCopy()` , `handleCut()`
[ ]	4.9	— 3 protection levels: basic/medium/aggressive	Medium	`protectionLevel` prop
[ ]	4.10	— Touch event interception	Low	`handleTouchEnd()`

5. Chapters & Transcripts

Status	#	Feature	Complexity	Source
[ ]	5.1	**Chapters** (from API metadata)	Medium	`chapters` in `authSlice.ts` , `setChapters()`
[ ]	5.2	Chapter normalization (API → player format)	Low	`normalizeChapters()`
[ ]	5.3	Chapter display in drawer	Medium	`DrawerTab: "chapters"`
[ ]	5.4	**Multi-language transcripts**	High	`transcriptMeta` , `transcriptsByLanguage`
[ ]	5.5	Transcript fetching (per-language URLs)	Medium	`fetchAllTranscripts()`
[ ]	5.6	Transcript segment sync (highlight current)	Medium	`TranscriptSegment` type

[ ]	5.7	Active transcript language switching	Low	`activeTranscriptLanguage`
[ ]	5.8	Transcript display in drawer	Medium	`DrawerTab: "transcript"`

6. Playlist / Queue System

Status	#	Feature	Complexity	Source
[ ]	6.1	**Queue management** (load, track list)	High	`queueSlice.ts` (5.2 KB)
[ ]	6.2	Next / Previous track	Medium	`nextTrack()` , `prevTrack()`
[ ]	6.3	Jump to specific track	Low	`playTrack()`
[ ]	6.4	**Shuffle** (Fisher-Yates, preserves current)	Medium	`toggleShuffle()`
[ ]	6.5	**Repeat modes** (off / all / one)	Medium	`setRepeatMode()` , `RepeatMode` type
[ ]	6.6	Autoplay next track	Low	`toggleAutoplay()`
[ ]	6.7	Loading state for next track	Low	`isLoadingNext`
[ ]	6.8	Playlist display in drawer	Medium	`DrawerTab: "playlist"`
[ ]	6.9	Per-track authentication (JIT auth)	High	`authRequired` on `AudioItem`
[ ]	6.10	Per-track metadata (chapters, transcripts)	Medium	`metaData` on `AudioItem`
[ ]	6.11	Per-track branding (logo, colors)	Medium	`brand` on `AudioItem`

7. Popup / Interactive System

Status	#	Feature	Complexity	Source
[ ]	7.1	**JSON-driven popup system**	Very High	`popupSlice.ts` (11.3 KB), `PopupRenderer.tsx` (17 KB)
[ ]	7.2	Time-based triggers (show at specific time)	Medium	`PopupTrigger` , `handleTriggerPopups()`

[ ]	7.3	Auto-pause on popup	Low	`pauseVideo` in trigger
[ ]	7.4	Show-only-once tracking	Low	`markPopupShown()`, `shownPopupIds`
[ ]	7.5	Auto-close with countdown	Medium	`autoCloseCountdown`, `decrementAutoCloseCountdown()`
[ ]	7.6	**Popup types:**		
[ ]	7.6a	— Action popup (CTA buttons)	Medium	`type: "action"`
[ ]	7.6b	— Rating popup (star rating)	Medium	`type: "rating"`
[ ]	7.6c	— Form popup (data collection)	High	`type: "form"`
[ ]	7.6d	— Multi-step form	High	`type: "multi-step-form"`
[ ]	7.6e	— Interactive elements (poll/quiz/reaction)	Very High	`type: "interactive"`
[ ]	7.6f	— Rich media popup (image/video/gif)	Medium	`type: "rich-media"`
[ ]	7.6g	— Countdown popup	Medium	`type: "countdown"`
[ ]	7.7	**Form system** (12 field types, validation)	High	`FormField`, `FormConfig`
[ ]	7.8	Multi-step form navigation	Medium	`nextStep()`, `prevStep()`, `saveStepData()`
[ ]	7.9	**Interactive elements:**		
[ ]	7.9a	— Poll (with results display)	Medium	`setPollResults()`, `incrementPollOption()`
[ ]	7.9b	— Quiz (correct answer, explanation)	High	`handleQuizCompletion()`
[ ]	7.9c	— Multi-question navigation	Medium	`MapsToQuestion()`, `goToNextQuestion()`
[ ]	7.9d	— Reaction buttons	Low	`type: "reaction"`

[ ]	7.10	Popup actions (redirect, resume, toast)	Medium	`handlePopupAction()` , `handleRedirect()`
[ ]	7.11	Custom popup styling (per-popup + global)	Medium	`StyleConfig` , `globalStyles` , `themes`
[ ]	7.12	Popup positioning (9 positions + custom)	Medium	`position` config
[ ]	7.13	Form data submission to API endpoint	Medium	`submitFormData` async thunk

8. Theming & Branding

Status	#	Feature	Complexity	Source
[ ]	8.1	**Dynamic theme from primary color**	Medium	`generateThemeFromPrimary()`
[ ]	8.2	Full theme generation (10+ CSS vars)	Medium	`generateTheme()`
[ ]	8.3	Theme styles application	Low	`getThemeStyles()`
[ ]	8.4	Contrast-aware text color	Low	`hasGoodContrast()` , `getReadableTextColor()`
[ ]	8.5	Hex color validation	Low	`isValidHexColor()`
[ ]	8.6	Custom primary/secondary/text colors	Low	Props on `SimpleAudioPlayer`
[ ]	8.7	**Watermark** display	Low	`watermark` prop
[ ]	8.8	Brand logo/colors per track (playlist)	Medium	`brand` on `AudioItem`

9. Analytics & Telemetry

Status	#	Feature	Complexity	Source
[ ]	9.1	**Event logging system**	High	`use-audio-analytics.ts` (6 KB)
[ ]	9.2	Analytics payload (audio + auth + context)	Medium	`AnalyticsPayload` interface

[ ]	9.3	Session ID generation	Low	`generateSessionId()`
[ ]	9.4	Audio ID extraction from URLs	Low	`extractAudioIdFromUrl()`
[ ]	9.5	Device context (mobile, orientation, battery)	Medium	`DeviceContext`
[ ]	9.6	Auth context in events	Low	`AuthContext`

10. Google Cast (Chromecast)

Status	#	Feature	Complexity	Source
[ ]	10.1	**Chromecast SDK integration**	Very High	`use-google-cast.ts` (14.3 KB)
[ ]	10.2	Cast button + device picker	Medium	`CastButton.tsx` (10 KB)
[ ]	10.3	Remote playback control	High	`UseGoogleCastResult`
[ ]	10.4	Cast session management	High	`initCast()`
[ ]	10.5	Remote state sync	Medium	`remoteCurrentTime` , `remoteIsPlaying`
[ ]	10.6	Media metadata on Cast receiver	Low	`CastMediaMetadata`

11. Embed / iframe System

Status	#	Feature	Complexity	Source
[ ]	11.1	Embed mode (minimal UI, no nav)	Medium	`embedMode` prop
[ ]	11.2	**PostMessage SDK** (parent ↔ iframe)	High	`use-audio-sdk.ts` (17.3 KB)
[ ]	11.3	PostMessage events (play, pause, etc.)	Medium	`onPlay()` , `onPause()` etc.
[ ]	11.4	Embed bounds reporting	Low	`embedbounds` message
[ ]	11.5	`<script>` tag embed loader	Medium	`hp-audio-player-embed.js`
[ ]	11.6	**Web Component** embed	High	`player.js`

12. Content Protection & DRM

Status	#	Feature	Complexity	Source
[ ]	12.1	Right-click disabled	Low	`use-content-protection.ts`
[ ]	12.2	Copy/cut prevention	Low	clipboard event handlers
[ ]	12.3	DevTools detection	Medium	`detectDevTools()`
[ ]	12.4	Touch protection (long-press)	Low	`handleTouchEnd()`
[ ]	12.5	Violation callback	Low	`onViolation` callback

13. Thumbnails & Media

Status	#	Feature	Complexity	Source
[ ]	13.1	**Thumbnail generation** (dynamic)	Medium	`use-simple-thumbnails.ts` (5.8 KB)
[ ]	13.2	Fallback thumbnail handling	Low	`thumbnail` prop fallback chain
[ ]	13.3	Album art / cover display	Low	Expanded player UI

14. State Management (Redux)

Status	#	Feature	Complexity	Source
[ ]	14.1	**5 Redux slices** (player, UI, auth, etc.)	High	`lib/store/`
[ ]	14.2	High-freq update optimization (refs)	Medium	`use-audio-player-redux.ts`
[ ]	14.3	Redux bridge hook	High	`useAudioPlayerRedux()` (22.5 KB)

15. Media Session API

Status	#	Feature	Complexity	Source
[ ]	15.1	**Browser media session** (lock screen)	Medium	`use-media-session.ts` (4.2 KB)

[ ]	15.2	Media metadata (title, artist, artwork)	Low	MediaSession API
[ ]	15.3	Action handlers (play/pause/seek)	Medium	MediaSession action handlers

16. Toast Notifications

Status	#	Feature	Complexity	Source
[ ]	16.1	Toast system (success/error/info)	Medium	`use-toast.ts` (3.9 KB)
[ ]	16.2	Toast on popup actions	Low	`toast` config in popup actions

Migration Priority Summary

P0 — Must Have (Core Functionality)

- Core Playback (1.1–1.12)
- UI Modes (2.1–2.4)
- Key Interaction (3.1–3.3)
- Auth & Security (4.1–4.5)
- Embed System (11.1–11.6)

P1 — Important (Product Definition)

- Chapters (5.x)
- Playlist/Queue (6.x)
- Theming (8.x)
- Media Session (15.x)

P2 — Valuable (Power Features)

- Popups/Interactive (7.x)
- Analytics (9.x)
- Chromecast (10.x)